

Thai Student Association Website GitHub Pages Deploy & Week 38 Specs

Working session transcript

DATE	Monday, 14 September 2026
REPOSITORY	campuscoderscrew/tsa-website
PROJECT	CCC Work — Campus Coders Crew
PARTICIPANTS	Brennen Le · Claude (Opus 5)
OUTCOME	GitHub Pages deployment configured and build-verified; 15 spec documents authored

1. The request

BRENNEN

Thai Student association website (tsa-website): Please update the build process so it can deploy to github pages.

Here are the TSA specs:

Club: Thai Student Association · umdtsa@gmail.com · Alison Wang (awang527@terpmail.umd.edu)

"Our club is a fun and welcoming organization focused on celebrating Thai culture. We would like to portray a festive spirit that is inviting to everyone. We would like to portray this mood in a clean way (colorful but not overwhelming) and feature elements of iconic Thai symbols within the website."

Colors: The colors of the Thai flag (Dark blue, Crimson Red, White) along with Gold and Deep Green

Theme: Thai Culture

Tone: Both! A fun but professional and clean look

Wanted: Board Member Gallery · Archive of Previous Board Members & Events · Events Section · About Section · Links to Other Platforms · A gallery for all events · A calendar

Attached is a pdf of some of TSA's instagram posts. The current website looks good, but it looks kind of basic. Please identify areas of improvement for the website to help make the website look more attuned to TSA's branding. **DO NOT make the changes;** instead, please create product specifications in the `product_specs/W38` folder...

Make each spec about 1 hour to complete. For changes that take more time, create multiple specs and chain them together. Include Objective, Spec title, Assigned Developers, status... Organize it like a class project, where the developer is given specs on what to do but they have to implement it themselves.

6 developers will be working simultaneously; some developers will be pair programming together. Expect 3–4 specs to be completed simultaneously. Identify a couple of features that they can work on this week. The availability of each developer is 1–5 hours per week. For features not worked on this week, create a document under `product_specs` called `features_TODO`. These specs will be used by college students in a web development club.

2. Investigation

Before writing anything, I read the attached Instagram PDF and worked through the repository on Brennen's machine — the full component tree, the Tailwind theme, both page components, the router setup, the build config, and the git history. I also checked a sibling repo (`CCC-Website`) to see how the club already deploys.

What the Instagram showed

TSA's actual visual language turned out to be much more defined than the website suggests:

- **Elegant high-contrast serif displays** on their headline posts — Songkran, Spring Picnic, Board Applications, Valentine's GBM
- **Cut-out collage lettering** and torn-paper tape labels for board member introductions
- **Lai thai / kranok border patterns** framing their board application posts
- **Temple, prang and chedi silhouettes** in their story highlight covers
- **Board cards** showing role set vertically in caps, then name, major, hobbies and a favourite snack
- Constant collaboration with sibling orgs — NSA, MCA, KSA, VSA

What the codebase actually contained

FINDING	DETAIL
Hardcoded colours	66 hex values, 23 distinct, across 13 files
Colour drift	Three different reds and four different navies, with no rule for which is correct
Design tokens	A full shadcn token set — every value greyscale. Not one brand colour.
Deep green	One of five colours TSA asked for. Appears nowhere.
Thai motifs	Zero. Decoration is generic blurred blobs and an abstract squiggle.
Placeholder copy	The word "Placeholder" is visible on the live site, four times
Hero avatars	Unsplash stock photos of strangers, labelled "Community"
Footer contact	<code>hello@tsa.club</code> , <code>href="#"</code> , socials pointing at platform homepages
Alt text	<code>alt=""</code> on every board and gallery photo
Events carousel	Drag-only — no keyboard access at all
Logo	A JPEG, so it renders as an opaque box on the dark hero and navy footer
Template leftovers	<code>StaggeredMenu</code> still ships purple <code>#5227FF</code> and <code>#ff0000</code> defaults
Dead code	<code>App.css</code> , <code>lotus_flower.png</code> , root <code>thai.jpg</code> , unused <code>@gsap/react</code>

MOST SERIOUS FINDING

The upcoming-events carousel can only be advanced by dragging. There is no keyboard alternative of any kind, which means a keyboard-only visitor cannot reach the second or third event. This was logged in the backlog rather than specced — it arguably deserves to jump the queue.

3. Three decisions put to Brennen

Three choices would have changed what got built, so rather than guessing I asked.

QUESTION	CHOSEN	CONSEQUENCE
How should the site deploy?	GitHub Actions workflow	Students merge to <code>main</code> and it publishes. No local deploy step, no <code>gh-pages</code> branch to get wrong.
Custom domain or default URL?	Default project URL	<code>campuscoderscrew.github.io/tsa-website/</code> — a sub-path, so <code>base</code> , router <code>basename</code> and a 404 fallback all have to be configured together.
How to fill "Assigned Developers"?	Leave unassigned	Every spec reads <i>Unassigned</i> — <i>assign at standup</i> , with a solo-or-pair recommendation and a reason.

4. The deployment work

A project site lives at a sub-path rather than a domain root, which breaks three things at once. All three had to be fixed together.

Vite base path

```
// vite.config.ts – overridable, so a custom domain later is a one-line change
const base = process.env.VITE_BASE ?? "/tsa-website/";

export default defineConfig({
  base,
  plugins: [react(), tailwindcss()],
  resolve: { alias: { "@": path.resolve(__dirname, "./src") } },
});
```

Router basename

```
// src/main.tsx – strips the prefix before route matching
const basename = import.meta.env.BASE_URL;

<BrowserRouter basename={basename}>
  <App />
</BrowserRouter>
```

SPA fallback

GitHub Pages has no rewrite rule, so a hard refresh on `/tsa-website/gallery` is a request for a file that does not exist. Copying `index.html` to `404.html` at build time lets the app boot and resolve the route client-side.

```
// scripts/copy-404.mjs, wired into the build script
"build": "tsc -b && vite build && node scripts/copy-404.mjs"
```

The workflow

`.github/workflows/deploy.yml` builds on push to `main` and publishes via `actions/deploy-pages`, with a `concurrency` group so two deploys can never race.

TWO THINGS STILL NEED A HUMAN

1. The workflow file could not be written to disk remotely — `.github/workflows/` is a protected path, by design. It was delivered through the chat instead and needs saving by hand.
2. A one-time repo setting is required or the workflow fails on its final step: **Settings** → **Pages** → **Source** = "GitHub Actions", not "Deploy from a branch".

Verification

Midway through, the device shell stopped mounting the project folder — a Windows update from 8 September. Rather than assume the build worked, I cloned the repository fresh in the cloud workspace, applied the changes, and ran the real build.

```
✓ 2197 modules transformed

dist/index.html          0.98 kB | gzip:  0.49 kB
dist/assets/index-R73aDy-C.css  64.13 kB | gzip: 11.31 kB
dist/assets/index-D7xcQjGV.js  577.83 kB | gzip: 198.71 kB

✓ built in 4.59s
copy-404: wrote dist/404.html (SPA fallback).
```

Then checked the output rather than trusting it: every asset in `dist/index.html` resolves under `/tsa-website/`, `404.html` is byte-identical to `index.html`, and no bare-root asset reference survives in the bundle. The existing components were already using `import.meta.env.BASE_URL` correctly, which is why this was a small change.

WORTH KNOWING

Vite rewrites root-relative paths inside `index.html` (`/logo.jpg` becomes `/tsa-website/logo.jpg`) but does *not* do this inside `.tsx` files. In components, always use `${import.meta.env.BASE_URL}foo.jpg`. This asymmetry is the most likely way someone ships a broken deploy.

5. The specifications

Twelve specs at roughly one hour each, chained where the work is longer, plus an index, a template and a backlog. Specs state *what* and *why* and never the code — the students implement it.

The parallelism problem

Six developers in one small repository is a merge-conflict machine. The specs are grouped into six **tracks**, and each track is deliberately assigned to a disjoint set of files. One file — `src/index.css` — is read by several tracks, so Track A owns it outright and everyone else requests changes rather than making them.

Feature 1 — Brand Identity System

SPEC	TITLE	TRACK	DEPENDS ON
W38.1.1	Define the TSA color token set	A	—
W38.1.2	Migrate hardcoded hex colors to tokens	A	W38.1.1
W38.1.3	Add a display typeface and type scale	A	W38.1.1
W38.2.1	Build the Thai ornament component kit	B	—
W38.2.2	Apply ornaments to section headers	B	W38.2.1
W38.5	Transparent logo, favicon set, social preview	E	—

Feature 2 — Board Member Showcase

SPEC	TITLE	TRACK	DEPENDS ON
W38.3.1	Create the board member data model	C	—
W38.3.2	Redesign the board card with name and role	C	W38.3.1
W38.3.3	Add the board member detail overlay	C	W38.3.2

Feature 3 — Gallery and Contact Polish

SPEC	TITLE	TRACK	DEPENDS ON
W38.4.1	Rethemed gallery page	D	W38.1.1
W38.4.2	Gallery captions, alt text, and grouping	D	W38.4.1
W38.6	Real contact details and social links	F	—

Suggested allocation

UNIT	COMPOSITION	SPECS	HOURS
Track A	Pair — sets conventions everyone inherits	W38.1.1 → 1.2 → 1.3	3
Track B	Solo — self-contained, mostly new files	W38.2.1 → 2.2	2
Track C	Pair — longest chain, fiddliest interaction work	W38.3.1 → 3.2 → 3.3	3
Track D	Solo	W38.4.1 → 4.2	2
Track E	Solo — good for the least available developer	W38.5	1
Track F	Solo — good for the least available developer	W38.6	1

The palette

W38.1.1 reconciles the three reds and four navies into ramps and adds the missing green. The two flag colours are fixed by Thailand's 2017 royal decree and must stay exact — they are the ones a Thai student recognises on sight.

--tsa-navy-900	#1B1638	--tsa-red-700	#7A1028	--tsa-gold-500	#E8C547
--tsa-navy-700	#241D4F	--tsa-red-600	#9B1B30	--tsa-gold-300	#F3DE97
--tsa-navy-500	#2D2A4A	--tsa-red-500	#A51931	--tsa-green-700	#1F5D4C
--tsa-cream-100	#FFFFFF0	--tsa-sand-100	#F0EBE6	★ = official Thai flag colour	

Spec anatomy

Every spec carries the same fields, so a developer always knows where to look: **Objective** · **Background** · **Files You'll Touch** · **Specification** · **Acceptance Criteria** · **Out of Scope** · **Hints** · **Definition of Done**, plus a metadata table with status, pairing recommendation, dependencies and branch name.

Two fields do unusual work. *Out of Scope* exists to stop a one-hour task growing into three during review. *Hints* points at the concept to look up — `:focus-visible`, `preserveAspectRatio`, `@media (hover: hover)` — without giving the answer, which is the part that makes these teaching documents rather than tickets.

6. Checking my own work

After drafting the specs I went back and verified every factual claim in them against the actual code, on the grounds that a spec confidently asserting something false will cost a student an hour of confusion. Three claims were wrong and were corrected before delivery.

SPEC	WHAT I HAD WRITTEN	WHAT WAS ACTUALLY TRUE
W38.5	"The favicon is 404ing in production right now" — because <code>href="/logo.jpg"</code> is root-relative.	It isn't. Vite rewrites root-relative paths in <code>index.html</code> at build time. Confirmed by inspecting <code>dist/</code> . Rewritten to explain the real asymmetry between <code>index.html</code> and <code>.tsx</code> files.
W38.1.2	"You should find roughly twenty occurrences."	66 occurrences of 23 distinct values. Replaced the guess with a measured per-file table, and added two colours I had missed: a stray fourth navy <code>#1C2B5A</code> , and the template purple still sitting in <code>StaggeredMenu</code> 's defaults.
W38.4.1	"The transition may have colors baked in" — vague, and wrong about which.	It has a specific crimson <code>#9B1B30</code> stroke. Named it, so whoever rethemes the gallery knows what will sweep across their new background.

Several other claims were checked and confirmed: `App.css` is imported nowhere and is safe to delete; `lotus_flower.png` and the root `thai.jpg` are referenced nowhere; `useGSAP` is never called despite `@gsap/react` being installed; Epilogue is loaded site-wide and used on exactly one line of text.

7. What was delivered

FILE	PURPOSE
<code>vite.config.ts</code>	Sub-path base, overridable via <code>VITE_BASE</code>
<code>src/main.tsx</code>	Router basename
<code>scripts/copy-404.mjs</code>	SPA deep-link fallback
<code>package.json</code>	Build script wired to the fallback
<code>README.md</code>	Rewritten — deploy guide, asset rules, troubleshooting table, project layout
<code>.github/workflows/deploy.yml</code>	Actions workflow — <i>delivered via chat, needs manual save</i>
<code>product_specs/README.md</code>	Index, feature grouping, allocation table, process
<code>product_specs/TEMPLATE.md</code>	Spec format for future weeks
<code>product_specs/W38/ ×12</code>	The week's specifications
<code>product_specs/features_TODO.md</code>	Unspec'ed backlog, four priority tiers

Twenty of the twenty-one files were written directly into the repository. The workflow file is the exception noted above.

The backlog

`features_TODO.md` covers everything identified but not scheduled, in four tiers: features TSA explicitly asked for and does not have (calendar, board and event archives, full event gallery, lightbox, join flow); content that is currently fake; quality work (accessibility audit, performance, SEO, the undecided dark mode, Thai script support); and nice-to-haves.

It closes with eight questions only the club can answer — who maintains the events calendar after handoff, where event photos should live, whether they want a contact form, which social accounts actually exist, whether the logo exists as a vector. Several of the highest-priority features are blocked on those answers rather than on engineering time.

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